Lung as the Gateway for Environmental Exposures in Pulmonary and Cardiovascular Disease: Research Gaps and Opportunities; NHLBI/NIH Workshop Report

Emily P. Brigham, M.D., M.H.S., University of British Columbia, Canada
Beena G. Sood, M.D., M.S., National Heart, Lung, and Blood Institute (NHLBI)
Jacqui M Marzec, M.S., National Institute of Environmental Health Sciences (NIEHS)
Taylor A. Poor, M.D., Ph.D. National Institute of Allergy and Infectious Diseases
Jésus A. Araujo, M.D., Ph.D., M.Sc., David Geffen School of Medicine at University of California, Los Angeles
John R. Balmes, M.D., University of California, Berkeley School of Public Health
Hasan Bayram, M.D., Koc University Hospital, Turkey
Kambez Benam, D.Phil., University of Pittsburgh
Christopher Carlsten, M.D., M.P.H., University of British Columbia, Canada
Ulrike Gehring, Ph.D., Utrecht University, Netherlands
Gillian C. Goobie, M.D., Ph.D., University of British Columbia, Canada
Anna Hansell, M.P.H., Ph.D., MRCP, FFPH, University of Leicester, United Kingdom
Yvonne Huang, M.D., University of Michigan
Salik Hussain, D.V.M., Ph.D., Health Sciences, West Virginia University
Ilona Jaspers, Ph.D., University of North Carolina School of Medicine
Kirsten Koehler, Ph.D., Johns Hopkins University
Alison Lee, M.D., M.S., Icahn School of Medicine at Mount Sinai
Meredith McCormack, M.D., M.H.S., Johns Hopkins University
Kari C. Nadeau, M.D., Ph.D., Harvard T.H. Chan School of Public Health
Meghan E. Rebuli, Ph.D., University of North Carolina School of Medicine
Stacie Reveles, Cystic Fibrosis Research Institute
Coralynn Sack, M.D., M.P.H., University of Washington
Chi-Ren Shyu, Ph.D., University of Missouri Institute of Data Science and Informatics
Valérie Siroux, Ph.D., Inserm, France
Brent Stephens, Ph.D., Illinois Institute of Technology
Ranu Surolia, Ph.D., The University of Alabama at Birmingham
Rhonda D. Szczesniak, Ph.D., University of Cincinnati
Rajesh Vedanthan, M.D., M.P.H., New York University Grossman School of Medicine
Rosalind Wright, M.D., M.P.H., Icahn School of Medicine at Mount Sinai
*Nadia Hansel, M.D., M.P.H., Johns Hopkins University
*Mary B. Rice, M.D., M.P.H., Harvard Medical School

*co-last author

Corresponding Author: Emily Brigham, M.D., M.H.S.
University of British Columbia, Canada, Centre for Lung Health
7th Floor, Gordon and Leslie Diamond Health Care Centre
2775 Laurel Street, Vancouver, BC, Canada V5Z 1M9

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Abstract

The National Heart, Lung, and Blood Institute (NHLBI)-sponsored workshop, *Lung as the Gateway for Environmental Exposures in Pulmonary and Cardiovascular Disease*, convened a multidisciplinary group of international experts to discuss the unique role of the lung as the interface with the environment, and the cardiopulmonary consequences of inhalational exposures. Increasing frequency and intensity of natural disasters and extreme weather events are resulting in increasing environmental exposures that affect cardiopulmonary health. Emphasizing the critical importance of enhanced understanding and protection, workshop participants heard from the mother of a child with lung disease who lived through the considerable practical and emotional challenges of wildfires and evacuations. Discussion focused on evidence indicating the health impacts of exposures across the lifespan, as well as the mechanisms underlying links between environmental exposures and pulmonary conditions (e.g., inflammation, oxidative stress, DNA damage, and microbiota dysregulation). Participants described state-of-the-art tools and technologies, identified critical scientific gaps and challenges, and highlighted opportunities for interventions. Looking to the future, interdisciplinary longitudinal research focused on complex air pollution mixtures and their interaction with physical conditions (e.g., extreme heat and humidity) is needed to help address the cardiopulmonary health consequences of environmental exposures and natural disasters. Integrated approaches—such as systems biology; multi-omics; epigenetics; gene-environment studies; implementation science; geospatial modeling; statistical and machine learning and artificial intelligence analyses of open-source data—are expected to complement *in vivo* and *in vitro* methods to advance the field, spur innovation, and guide the development and implementation of interventions and policies.

Introduction

Natural disasters and extreme weather events (e.g. wildfires, dust storms, extreme heat, earthquakes, floods) impact cardiopulmonary health. The lung is directly affected and also serves as a primary line of defense for the rest of the body against inhaled exposures. Impacts of exposure have the potential to accumulate over the lifespan, influencing lung development, lung function, and pulmonary symptom/disease risk [1,2]. Effects extend beyond the lung, and include cardiometabolic health effects (e.g., atherosclerosis and ischemic events) and all-cause mortality [3–7]. The links between environmental exposures and an array of adverse health effects highlight the importance of protective actions, including interventions to reduce exposure at the individual, local, and national levels [2].

Given increasing environmental exposures and their associations with adverse effects on cardiopulmonary health, the National Heart, Lung, and Blood Institute collaborated with the National Institute of Environmental Health Sciences and the National Institute of Allergy and Infectious Diseases to convene a 2-day virtual workshop, *Lung as the Gateway for Environmental Exposures in Pulmonary and Cardiovascular Disease* (January 26–27, 2026). Interdisciplinary experts—including basic, clinical, and implementation scientists; clinicians; epidemiologists; data science specialists, and an individual with lived experience—from the United States, Canada, France, Netherlands, United Kingdom, and Turkey participated. Presenters discussed the unique role of the lung as the interface with the environment, and the pulmonary and cardiovascular consequences of inhalational and heat exposures to identify critical research gaps, challenges, and opportunities to advance research and interventions to mitigate potential health effects. Specific areas of focus included:

- Cardiopulmonary effects of environmental exposures across the life course
- Mechanisms of inhaled environmental exposures and cardiopulmonary health
- Emerging tools and technologies for environmental exposure assessment
- Interventions to mitigate cardiopulmonary effects of environmental exposures
- Patient and caregiver experience across events

Setting the Stage—The Lung as the Gateway for Environmental Exposures

Respiration, and the process of gas exchange, is often considered the primary function of the lung. The alveolar surface area serving this purpose is commonly estimated at ~70m^2 in an adult[8], which does not include the vast but inconsistently quantified surface area of the branching airways within the lung. An average adult exchanges approximately 6L of air (and exposures therein) per minute at rest during normal respiration[9], approaching 10,000 L of air (and components therein) interacting with this surface area daily.

Beyond respiration, the lung senses and responds to various inhaled exposures (e.g., pollutants and allergens) under different conditions (e.g., moisture, pressure, and temperature) through immune and neuronal systems[10]. Inhaled pollutants, interacting with the surface area of the respiratory system, trigger inflammation, oxidative stress, DNA damage, telomere shortening, epigenetic instability, microbiome changes, and immune dysregulation (**Figure 1**)[11–14]. Early life exposures (from the prenatal to early childhood periods) program lung function and affect health across the lifespan[15]. Children are more susceptible to the adverse effects of inhaled exposures on respiratory health due to development and growth of immune and organ systems, higher levels of physical activity, and a higher respiratory rate combined with lower body mass (for inhaled

pollutants, this results in a higher effective dose compared to adults at a given concentration)[16]. Natural disasters and extreme weather events (e.g. wildfires, heat waves, earthquakes[17], and dust storms) elevate exposures (e.g. particulates/dust, mold, pollen, gases, toxicants) and are associated with increased cardiopulmonary morbidity and mortality [18,19]. (**Table 1**) Ongoing environmental exposures have the potential to accumulate across the lifespan, impacting peak lung function attainment, lung function trajectory, and risk of respiratory disease and attributable mortality[20]. Inhaled exposures are further associated with cardiovascular effects including hypertension, arrhythmias, and ischemic events, and mortality from cardiovascular causes[21,22].

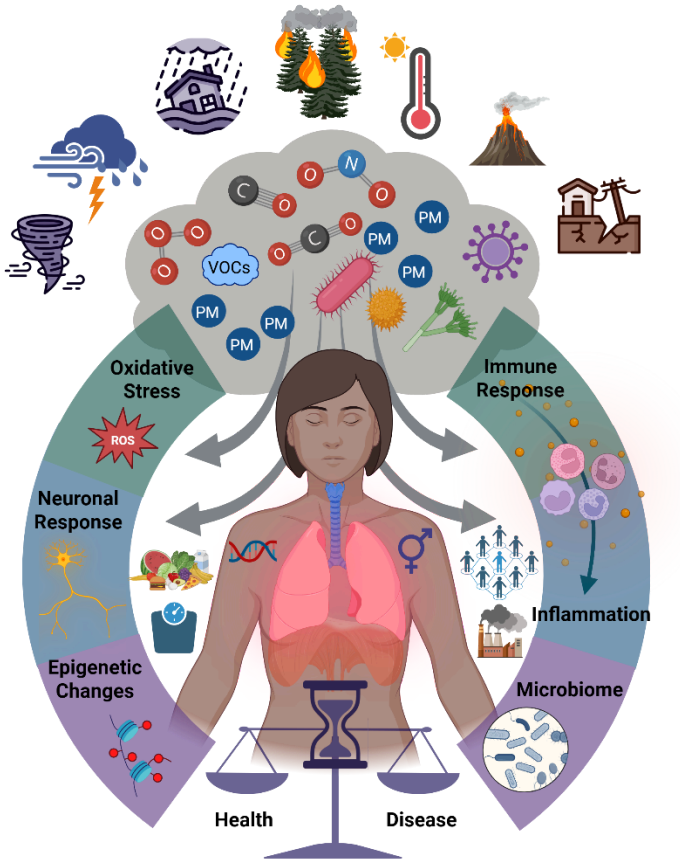


Figure 1. Lung as the Gateway. Inhaled exposures from extreme weather events contribute to the release and distribution of airborne pollutants (particulate, gaseous), infectious agents (fungal, bacterial), and allergens (i.e. pollen). Respiratory exposure activates a variety of biologic mechanisms (immune response and inflammation, reactive species production, neuronal response, epigenetic changes, and shifts in the respiratory microbiome), modified by intrinsic (including sex, genetics, and age) and extrinsic (social context, weight/metabolic health, diet, other pollutants) characteristics, ultimately contributing to the balance between health and disease in cardiorespiratory systems and beyond. Image: Biorender.com, Flaticon.com

Table 1: Weather Events/Natural Disasters, Resultant Exposures, and Cardiopulmonary Effects

Weather Event/ Natural Disaster	***Exposure***	***Respiratory Effects***	***Cardiovascular Effects***
Extreme Heat	High temperature Ground level ozone Secondary PM Stagnant Air	Exacerbations Heat stress	Arrhythmia Hyper/Hypotension Ischemia Heart failure
Extreme Cold	Low temperature Low absolute humidity	Exacerbations Infection	Arrhythmia Ischemia Heart failure
Floods	Contaminated water Mold/fungal exposure	Asthma/allergic disease Infection Exacerbations	Arrhythmia Ischemia Heart failure
Cyclones, Hurricanes and Storms	Contaminated water Mold/fungal exposure Allergens Dust/PM	Infection Exacerbations Thunderstorm asthma	Hypertension Ischemia Heart failure/ Cardiomyopathy
Wildfires and Smoke	Pathogens Particulates (↑$PM_{2.5}$) PAH Chemicals and Toxicants	Exacerbations Infection ↓Lung function	Arrhythmia Hypertension Ischemia
Volcanic Eruptions	SO_2, CO_2, HCl, HF Particulates/Dust Volcanic ash (crystalline silica and other toxic particles)	Exacerbations Infection Acute respiratory distress	Arrhythmia Hypertension Ischemia Heart failure
Earthquakes	$PM_{2.5}$, silica, asbestos, radon gas, CO	Respiratory trauma, acute and chronic respiratory diseases, infections, and increased risk of lung cancer	Hypertension Ischemia Heart failure/ Cardiomyopathy

PM_{10}: particles with an aerodynamic diameter <10 μm; $PM_{2.5}$: particles with an aerodynamic diameter <2.5 μm; UFPs: ultrafine particles (<0.1 μm); PAH: polycyclic aromatic hydrocarbons; SO2: sulfur dioxide; CO2: carbon dioxide; HCl: hydrogen chloride; HF: hydrogen fluoride; CO: carbon monoxide. Respiratory effects adapted from Annesi-Maesano and colleagues[23]. Additional references[24–29]

Emphasis on Inhaled Pollutants

Presentation of material and discussion surrounding air pollution exposures was prominent, and additional background is therefore provided here for contextualization. Air pollution represents a complex mixture inclusive of particulate matter (PM) and gaseous components (including carbon monoxide (CO), ozone (O_3), nitrogen dioxide (NO_2), and sulfur dioxide (SO_2)). While gaseous components are fixed in composition, particulate matter contains a variable mixture of compounds (which may include metals such as lead and other elements with recognized health effects) and adsorbed toxicants and chemicals at its surface. Toxicity is dependent on source, composition, atmospheric transformations, and photochemical aging of components. PM is further categorized based on size fraction: PM_{10} (<10 μm in diameter), $PM_{2.5}$ (fine PM, <2.5 μm in diameter) and $PM_{0.1}$ (ultrafine PM, <0.1 μm in diameter). $PM_{2.5}$ is the most studied, demonstrating consistent evidence of human health effects and ability to travel to the small

airways.[30] Ultrafine particles, the smallest size fraction of $PM_{2.5}$ have further been found to translocate across the alveolar membrane and into the bloodstream[31,32].

Forward Focus

Workshop participants were asked to identify knowledge gaps and opportunities for discovery to inform research priorities moving forward. While reviewing evidence and work in progress, participants across the workshop highlighted a need for interdisciplinary research, integrated methodologic approaches, and knowledge translation (**Table 2**).

Table 2. Research gaps and opportunities in environmental exposures and cardiopulmonary health.

Area	***Research Gaps and Opportunities***
Cross-Cutting Themes	• Interdisciplinary research teams and investigations • Attention to multimorbidity/diverse effects • Strength in crossing translational valleys (basic, cohort, population epi) for expanding discovery
Focus Area 1: Cardiopulmonary Effects of Environmental Exposures Across the Life Course	• Timing -Consideration of differential effects of exposure at varying age/developmental windows -Single events, intermittent vs. chronic exposures -Short and long-term exposures and effects • Location -Geographic representation • Toxicity -Source-specific -Composition/mixture-specific • Co-Exposure -Pollutant mixtures -Interaction with temperature, pollen, etc. • Health outcomes -Systemic manifestations/multimorbidity -Disease programming/progression • Precision medicine -Susceptibility/vulnerability/resiliency -Sex-stratified analyses (NIH SABV policy)
Focus Area 2: Mechanisms of Inhaled Environmental Exposures and Cardiopulmonary Health	• Susceptibility and resiliency -Genome wide association -Gene x environment interactions • Therapeutic target discovery -Locus-specific DNA methylation -NLXR2 • Pathways -Lung-Systemic

Area	***Research Gaps and Opportunities***
	-Gut-Lung -Gut-Vascular • Outcome-specific mechanisms -Mediating role of immune cells
Focus Area 3: Emerging Tools and Technologies for Environmental Exposures Research	• Low-cost environmental sensors - Improve accuracy/specificity - Reduce/optimize sensor energy requirements - High-resolution and hyper-local data • Biomarkers - Early life lung function: tidal breathing flow-volume loops (TBFVL), nitrogen multiple breath washout (N_2MBW), airwave oscillometry (AOS) • Wearable Technology • Data Opportunities - Open-source data, data sharing, and harmonization - Leverage relevant data from longitudinal cohorts - Electronic health records • Analytic opportunities - Distributed Lag Models (DLMs) and BDLMs - Scenario simulations - Clinical risk prediction - Machine learning/artificial intelligence: integration of multiple exposures, biomarkers, -omics, and health outcomes
Focus Area 4: Interventions to Mitigate Cardiopulmonary Effects of Environmental Exposures	• Dosing - Environmental change needed to see clinical effect - Duration of intervention - Pollutant dynamics • Susceptibility - Identification and testing - Stratification/Power • Prioritization - Pollutant/Source - Population - Outcomes • Generalizability versus customization - Geographic location - Housing/building characteristics • Adherence • Cost-effectiveness • Accountability and sustainability
Focus Area 5:	• Anticipatory action/intervention - Effective mental health support

Area	***Research Gaps and Opportunities***
Patient and Caregiver Experience Across Events	- Human connection - Advanced identification of challenges and proactive support ● Community engagement - Patients and family - Patient advocacy organizations

Patient and Caregiver Experience Across Events

Underscoring the critical importance of work in this field was a presentation by a mother of a child living with a chronic respiratory disease who experienced a wildfire event with her family. She described the continuous challenges they faced in their efforts to prioritize the health of their child, necessitating a stable home environment and access to medical facilities. Power outages impacted home medical equipment and medication storage/viability. Decisions regarding relocation and evacuation were recurrent. Getting to care appointments was obstructed by either smoke or, in one instance, a fire between their home and the clinic. All of this was exacerbated by the ongoing COVID pandemic also requiring isolation and complicating evacuation planning and contributing to shortages in medication supply on a larger scale. Permanent relocation was considered, but was neither affordable nor feasible. Exhaustion, fear, and guilt were prominent emotions.

Caring for a family member with a chronic cardiac or respiratory disease while living in an area affected by wildfires and evacuation orders is challenging and stressful, with adverse effects on mental health. In the absence of clear guidance from authorities, evacuation-related decision-making can be difficult emotionally and practically. Connecting with organizations that support patients can help researchers, care professionals, and public health and safety officials better engage with families and develop strategies for mitigating these impacts.

Cardiopulmonary Effects of Environmental Exposures Across the Life Course

Intrauterine, Early Postnatal, and Early Childhood Exposure—Environmental exposures in the perinatal period and during childhood and adolescence have been shown to adversely affect lung development, lung function trajectory, and increase the risk of respiratory or asthma symptoms[20]. Lung development begins in utero as early as 3-4 weeks post conception, followed by predictable stages of growth and development that have implications for lung health postnatally[33]. Indeed, the respiratory system is especially vulnerable to environmental pollutants over gestation and in the first 18-36 months of life[34–36], and sensitive windows for exposure have been discovered for several pollutants ($PM_{2.5}$ and nitrate, NO_3^-) with regards to child lung health outcomes[37–39]. Potential interactions with temperature, between pollutants in complex mixtures, and diverse geographic representation are underrepresented in research to date with regards to early life inhaled exposures. Lung function trajectories and failure to achieve peak lung growth can mark subsequent risk [40–42] (**Figure 2**), with some evidence suggesting that fixed growth patterns may be established as early as age 6 years,[43] emphasizing the urgency of research in this space.

Early-life exposure (pre-adulthood) to airborne pollutants has also been associated with higher risk for and prevalence of multiple cardiovascular health indicators, including higher blood pressure, dyslipidemia, atherosclerosis, and arrhythmia[44]. Risks of exposure extend to the prenatal period, where exposure to $PM_{2.5}$ has been linked to higher blood pressure in childhood [45,46], highlighting the need for multimorbidity considerations in health effects from early life exposure.

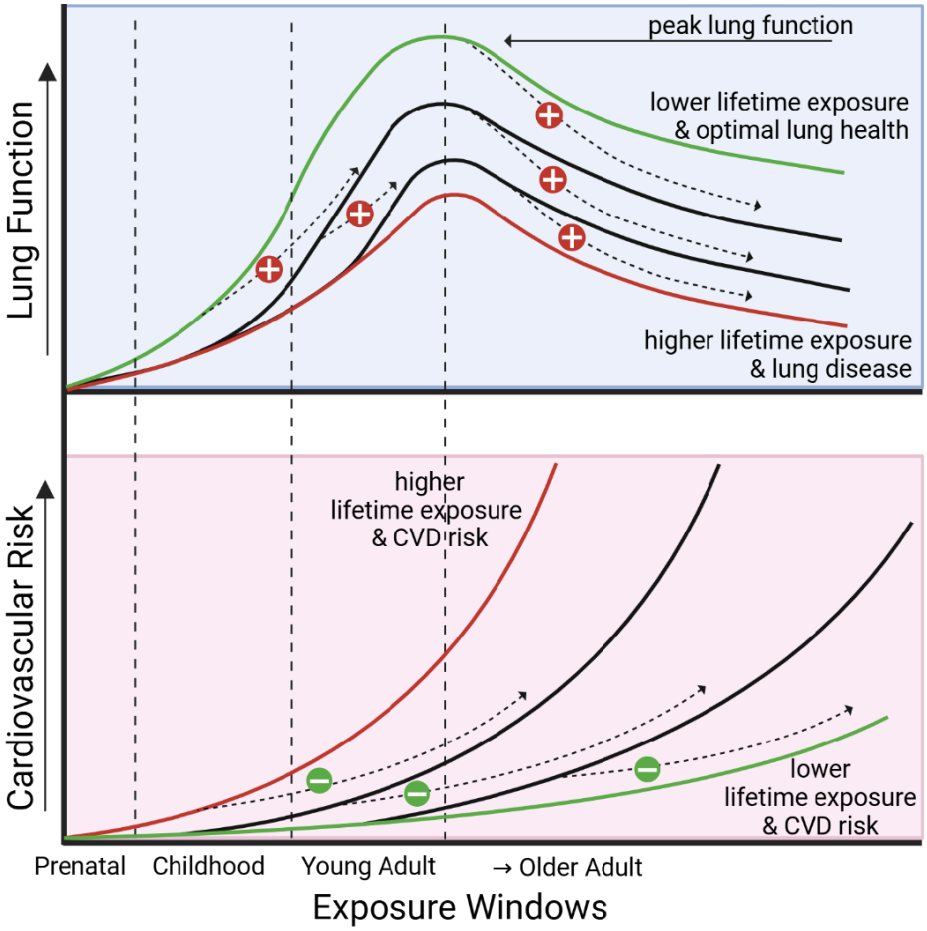


Figure 2. Airborne pollutant exposure across the lifecourse and impact on cardiorespiratory health; a schematic. Prenatal exposures with ongoing exposures throughout life carry the greatest potential for disease risk (red) while minimized exposure confers the greatest opportunity for health (green). Additional exposures (red + in respiratory figure or solid line transitions in cardiovascular figure) may alter trajectory favoring disease, while reductions in exposure (green – in cardiovascular figure or solid line transitions in respiratory figure) may preserve function or improve risk profile. The figure represents a simplified schematic; potential for additional transitions between states is plausible. Respiratory figure adapted from multiple works, including Fletcher and Peto[47], Lange and colleagues[40], Reyfman and colleagues[48], Liu and Kalhan[49]. Cardiovascular figure adapted from Zhang and colleagues[44]. Created in Biorender.com. CV= cardiovascular

Sex-Specific Effects— Research on sex differences in health conditions and their underlying biology has been significantly advanced by the introduction of the 2016 National Institutes of Health (NIH) sex as a biological variable (SABV) policy [50]. Studies have shown sex differences in the incidence, morbidity, mortality of asthma, COPD, pulmonary fibrosis, lung cancer, pulmonary hypertension, and other lung conditions[51], multiple of which are associated with or worsened by air pollutant exposures. Males and females show anatomical and physiological differences in the respiratory system that alter mechanical flow and lung function and may affect both deposition of inhaled exposures and the impact of inhaled environmental exposures on health. For example, women on average have lower lung volume, fewer alveolae, a smaller cross-sectional area of the trachea, and increased bronchial responsiveness[52]. Estrogen receptor expression and activation differs between sexes and across life stages with hormonal fluctuations in a variety of cell types within the lung,[53,54] dynamically regulating lung homeostasis and contributing to both protective and pathogenic responses and health outcomes[54]. Sex differences in toxicokinetics and pharmacokinetics (e.g., cytochrome P450 expression) may also underlie potential sex-specific responses to environmental exposures[55,56]. Sex differences in response to air pollution exposure include differential immune-related gene expression in the primary respiratory epithelial cells; in response to wood smoke exposure, men show elevated inflammatory gene expression, whereas women demonstrated suppressed host defense gene expression[57].

Other Susceptibility Factors— Individuals with pre-existing lung disease are also considered at higher risk in the setting of inhaled exposures and heat. As example, among patients with established fibrotic interstitial lung disease (fILD), higher concentrations of $PM_{2.5}$ are associated with increased mortality[58], and O_3 and NO_2 elevations associate with fILD exacerbation risk[59]. Exposure to $PM_{2.5}$ notably increases the risk for the development of idiopathic pulmonary fibrosis (IPF), a form of fILD[60]. The impacts of heat specific to patients with fILD are under-explored, however, given increased risk from heat waves for patients with underlying respiratory disease, patients with fILD are considered a susceptible subgroup[61,62]. Natural disasters also contribute to increased organic dust and mold exposures, both relevant to the risk of hypersensitivity pneumonitis (another form of ILD)[63,64]. Further, up to 40% of patients with fILD require supplemental oxygen[65], and are adversely impacted by service disruptions during natural disasters and extreme weather events.

As noted above regarding sex-differences, anatomic variation in airway morphology may alter the deposition, distribution, and retention of inhaled noxious exposures. Dysanapsis refers to a mismatch between airway tree caliber and lung parenchyma size[66,67], and is linked to a broad spectrum of diseases and poor outcomes across the lifespan—including morbidity in bronchopulmonary dysplasia[68] and asthma[69,70], accelerated lung function decline[71], COPD incidence and progression[72,73], and all-cause mortality[74]. Individuals with narrower airway trees have faster particle velocity, increased turbulent flow, and greater impaction of micrometer particles at both airway bifurcations and in alveolar regions[75]. Cross-sectional and longitudinal studies among individuals with chronic lung disease and in the general population show that smaller airway tree caliber is associated with increased adverse effects from indoor and outdoor air pollution exposure[76,77].

Patients with asthma, defined by airflow limitation and environmental sensitivity are susceptible to asthma-related morbidity and mortality associated with extreme weather events[78]. Exposure to

dust storms, wildfire smoke, heat, elevated airborne molds and bacteria due to hurricanes, and the lengthening of pollen season all contribute to increase morbidity and/or mortality among individuals living with asthma[78–84]. Notably, pollutant particles adhere to pollen grains and alter their shape in ways that may exacerbate airway irritation[85], potentially compounding risk. Among patients with asthma, women and children tend to demonstrate the highest sensitivity to extreme weather events[78].

A genome wide association study (GWAS) with approximately 300,000 unrelated European individuals from the UK Biobank points to potential genetic susceptibility to the effect of air pollution on lung function. Analyses identified seven new significant genetic signals modifying the lung function response to air pollution—mostly rare variants spread over chromosomes 1, 4, 5, 6, 7,19, and 21 (2 variants on chromosome 21), and one previously identified signal[86]. No significant interaction with a weighted genetic risk score[87] or several other commonly evaluated antioxidant genes[88] were found; replication requires access to datasets of even larger magnitude.

Source-Specific Effects-- Wildfires are an increasing source of air pollution nationally and globally, with smoke from wildfires travelling thousands of miles from the source and documented health impacts [89–91]. Wildfire smoke components include PM, CO, NO_2, polycyclic aromatic hydrocarbons, heavy metals, and volatile organic compounds[92,93]. As the smoke travels form the source and interacts with ultraviolet light and the atmosphere, secondary PM and O_3 are formed. Increasingly common, wildfires in the wildland-urban interface involve the burning of buildings and vehicles, which releases other toxic air contaminants (e.g., HCN, HCl, phosgene, toluene, styrene, and dioxins). Documented adverse respiratory health effects of wildfire smoke exposure include exacerbations of asthma and COPD, respiratory infections, and related healthcare utilization [89–91,94–96]. Wildland firefighters, recurrently exposed to wildfire smoke, demonstrate cross-shift and cross-season changes in lung function and inflammatory markers[97]. Based on effect estimates from ambient PM exposure, lifetime exposures of wildland firefighters to wildfire PM are predicted to significantly increase the risk of cardiovascular disease and lung cancer[98]; the relative toxicity of wildfire smoke as compared to other sources of PM is not established though some investigations suggest risk may be elevated[99,100].

Environmental heavy metals are a variable component of airborne particulate matter, entering the air as a result of both natural events, such as volcanic eruptions, wildfires, tsunamis, and earthquakes, and human activities, including smelting, welding, fossil fuel combustion, mining, burn pit fires, and cigarette smoking. Environmental metal exposures cause chronic lung diseases and worsen pre-existing conditions. Metals such as cadmium[101,102], mercury[103,104], arsenic[105,106], and lead[107] are associated with lung injury, inflammation, decreased lung function, and diseases.

***Extreme Heat*—**Following nearly two centuries of rising average global surface temperatures[108,109], extreme heat events have increased in frequency and duration[110,111], and some subgroups (e.g., patients at the extremes of age, and those with pre-existing respiratory and cardiovascular disease) are particularly susceptible. Other susceptibility factors include comorbidities, lack of air conditioning, poor housing conditions, and limited green space. Some communities, including those in urban heat islands characterized by high nighttime temperatures, may be more vulnerable to adverse health effects. A robust evidence base links heat exposure with increased risk of exacerbated cardiopulmonary conditions—such as prolonged device-detected atrial fibrillation episodes, arrhythmia, and heat-related emergency hospitalizations for respiratory disease among older adults [112–116]. Heat may also interact with air

pollution to compound negative effects on health (i.e. lung function, symptoms)[117,118]. Extreme heat may worsen air pollution risks by contributing to excess O_3 generation[119], which is associated with increased risk of emergency visits for asthma, allergies, and ear disorders among children [120].

Mechanisms of Environmental Exposures Affecting Cardiopulmonary Health

Inflammation and Innate/Adaptive Immune Responses—Air pollution, including as a result of or exacerbated by extreme weather events and natural disasters, contributes to inflammation in the lungs and the occurrence or aggravation of respiratory diseases[121–123]. Interaction of air pollutants (from dust and sandstorms, wildfires, and anthropogenic sources) with the respiratory epithelium triggers activation and migration of neutrophils, macrophages, and other cells of the innate immune system[122]. Observed and expected increases in pollen season length and/or pollen concentrations are expected to contribute to an increased severity and possibly increased prevalence of allergic disease via Th2 cell recruitment, which may be further modified by exposure to airborne pollutants[80,122,124–127].

Exposure to the heavy metal cadmium, released by natural events (wildfires, dust storms), resuspension of dust, combustion of tobacco smoke and fossil fuels, promotes persistent respiratory inflammation through dysfunctional efferocytosis (clearance of apoptotic cells by phagocytes), which contributes to emphysema and COPD[128,129]. Mechanistically, cadmium upregulates MerTK but simultaneously promotes its cleavage by ADAM17, generating sMer, which acts as a decoy receptor that hinders effective efferocytosis [130]. Animal research suggests that the nucleotide-binding oligomerization domain-like receptor NLRX1 plays an important role in the cellular response to the inhalation of O_3 and co-exposure to black carbon[131]. Expressed in both human and mouse lungs and implicated in COPD, NLRX1 is mostly found in macrophages and negatively regulates inflammatory responses [132]. This finding suggests that NLRX1 agonists could potentially counter air pollution-induced pulmonary inflammation and injury.

Oxidative Stress—Production of excessive reactive oxygen species (ROS) and subsequent oxidative stress modulates exposure-induced inflammation and contributes to DNA damage[121,122]. The oxidative potential (OP) of PM—i.e., its ability to generate ROS and trigger oxidative stress— serves as a potential biomarker for relative toxicity. Recent studies suggest that the OP of PM may be a more accurate predictor than mass concentration (alone) for respiratory health outcomes[133]. Additionally, epidemiological work indicates that prenatal exposure to PM with high OP can impair lung function in early childhood and affect maternal oxidative stress biomarkers[134,135]. Such mechanisms underpin pathophysiological changes—such as airway remodeling, fibrosis, and carcinogenesis in airways and alveoli, and atherosclerotic risk—ultimately resulting in greater risk for cardiopulmonary morbidity and mortality.

Epigenetics—The epigenome, or the collection of chemical changes to DNA that affect when, where, and how DNA is expressed, may play a role in mediating the health effects of environmental exposures, especially in the setting of chronic lung conditions. For example, epigenetic aging is accelerated among patients with fILD and mediates a portion of the relationship between $PM_{2.5}$ exposure and mortality [136,137]. Locus-specific DNA methylation analyses may help identify novel environmentally sensitive therapeutic targets [138].

***Microbiome*—**Environmental exposures directly (as a vector for exposure) or indirectly (via modification of innate immunity) affect regulation and composition of airway microbial communities[14,139]. For example, exposure to the mold *Aspergillus* (ubiquitous in the environment, detected in samples of indoor and outdoor air, and demonstrating increased allergenicity in the context of urban air pollution[140]) is associated with exacerbation of COPD [141]. Microbial communities may also modify host responses to air pollution[14,142]. Lower airway bacterial diversity is linked with COPD and a heightened inflammatory response to air pollution [139,142]. Other pollution-associated modifications of innate immune response include suppression of host defense[143]. *In vitro* work indicates that particulate matter increases viral load in human bronchial epithelial cells in—an effect worsened by inhaled corticosteroids[144]. Further, inhaled ultrafine particulates have been found to induce changes in the gut microbiota, influencing systemic oxidative stress with worsened atherosclerosis[145].

Emerging Tools and Technologies for Environmental Exposures Research

***Exposure Assessment and Linkage*—**Low-cost sensors offer an opportunity to increase the spatial resolution of regulatory monitoring networks[146], and can be applied to personal monitoring, evaluating indoor air quality, and the efficacy of interventions[147,148]. Calibration is an important consideration to improve accuracy[149,150], and dynamic calibration which incorporates spatial correlation can enable unified predictions for expanded spatial regions[151]. Iterative improvements in sensor accuracy and capabilities (i.e. pollutants measured, reduced device power requirements) can expand application of this technology to enhance discovery.

Satellite imagery and remote sensing products add to a research infrastructure underpinned by open geographic information system (GIS) data, and increased spatiotemporal resolution of exposure (via by machine learning and artificial intelligence) can facilitate hyper-local data access and improve precision of exposure-health associations and predictions[152]. Geospatial artificial intelligence (AI) is an emerging tool enabling the integration of multiscale environmental exposures, which can then be linked with clinical, genomic, and real-world health data[153]. Key resources supporting efforts in this area include the Geospatial Analytical Research Knowledgebase (GeoARK), which stores curated exposomes and allows queries of environmental exposures[154]. Linkages more broadly can be applied to identify novel subgroups[155] and spatial disparities[156]. Privacy-preserving linkages and governance are essential, as well as clear communication of bias/uncertainty, responsible use, and standards for reproducibility.

***Human-Relevant Preclinical Tools*—** An evolving landscape of *in vitro* and *ex vivo* methodologies have expanded the potential for mechanistic insights into exposure-health relationships. These include three-dimensional cell culture and physiologically-relevant models (e.g. spheroids, organoids, 3-dimensional bioprinting, precision cut lung slices, and Transwell inserts)[157]. Frontiers in organotypic systems representative of human structure and function include emulation of the biomechanical forces of airflow and vascular shear, blood-like flow, and enhanced organ-level complexity.

To address some of these gaps and to model microphysiological environments relevant to inhaled environmental exposures, researchers have developed specific organ-on-chip technologies. For example, the Human Lung Small Airway-on-a-Chip features an air lumen and blood channel and recapitulates patient-specific responses (e.g., smoke-induced oxidative stress and elevated IL-8) [158]. The Extracellular Matrix-embedded Microvasculature-on-a-Chip features endothelial cells

and fibroblasts. In this microphysiological system, researchers can control the nature of the fibroblasts (e.g., healthy or fibrotic) or can change blood vessel media to examine vascular pathology remodeling[159]. Though these tools can significantly enhance understanding of the effect of environmental exposures on human health and complement other research approaches, their inability to recreate organ-organ crosstalk and model organ systems is a significant limitation and an area in need of further innovation[160].

Bioinspired robotics platforms, engineered to mimic the actions involved in human breathing, can quantify the number and size distribution of particles in real-time with regards to inhaled exposures[161]. Further, laboratory combustion chambers (including quartz tube combustion systems) allow for chemical analysis of combustion mixture emissions. This model is especially valuable when materials combusted vary over space and time, such as during wildfire events, with measured composition and applied computational modelling supporting integrated chemical-biological profiling[162].

Interventions to Mitigate Cardiopulmonary Effects of Environmental Exposures

***Interventions to Reduce Indoor Air Pollution*—**Up to 90% of time daily is spent indoors [163], making the indoor environment an important target for intervention to reduce exposure. Indoor pollutants (e.g., PM and organic and inorganic gasses) are a threat to public health globally and originate from sources such as cook stoves, cleaning products, combustion, building materials, and outdoor air infiltration into the indoor environment[164]. Residential indoor air pollution is estimated to result in 5-14% of the annual non-communicable, non-psychiatric disease burden in the U.S[165]. Individuals with chronic conditions, including lung disease, are particularly susceptible to pollutant effects[166–168].

Four principles underlying good air quality include minimization of indoor emissions (i.e. via product standards, behavioral interventions), keeping buildings dry (avoid mold and mildew), ventilate well (remove indoor pollutants), and protect against outdoor pollutants (consider building envelopes, mechanical ventilation, and window openings)[169]. Government agencies recommend interventions to improve indoor air quality[170]; specific building operational details and use patterns are crucial considerations to optimize benefit. For example, central heating and cooling systems using smart thermostats are only run about 20% of the time[171] and few people operate kitchen exhaust ventilation fans (range hoods) regularly[172], providing opportunities for intervention.

Effective air cleaning may be a fifth principle. Household interventions include high-efficiency particulate air (HEPA) filters and cleaner-fuel stoves. Among people with allergic asthma, air filtration contributes to reduction in airway inflammation (as indicated by Fractional exhaled nitric oxide, FeNO) and improved quality of life.[173] Among people with COPD, HEPA plus carbon filter use decreases indoor levels of $PM_{2.5}$ and NO_2 and is associated with reduced rescue medication use and fewer exacerbations in symptoms [174]. This impact is attributable to reductions in indoor $PM_{2.5}$ [175]. Further, an environmental intervention (combined HEPA air cleaner receipt and motivational interviewing to support a home smoking ban and secondhand smoke reduction) demonstrated potential health benefits (improved symptoms) among current smokers with COPD[176]. Indoor air filtration has also demonstrated cardiovascular benefits in patients with COPD [177], and clinically important reductions in systolic blood pressure among individuals with

elevated SBP at baseline[178]. Addressing factors related to adherence was highlighted as important in both trials and future implementation[179]. An ongoing study (PURI-HF) is examining whether HEPA filters improve cardiovascular outcomes among patients with heart failure in three cities in India. Remaining questions in indoor air filtration trials include duration of intervention and degree of pollutant reduction needed to see effect, which pollutants are the priority, which individuals are most likely to benefit, and which strategies are transportable across settings versus which require customization to setting. Ongoing studies (APECS[180], a trial in US Veterans with COPD[181], a multi-Center environmental study in COPD: Clean Air[182], and others) will address some of these questions.

A study in Ghana found that providing women with cleaner fuel stoves during early pregnancy and through the child's first year of life reduced indoor air pollution exposures [183]. Additionally, children had better lung function and lower diastolic blood pressure at age 4 and lower risk for high blood pressure through age 9 compared with those in the control group[184–186]. Mothers in the intervention group also demonstrated better blood pressure through eight years postpartum, as compared to the control group[185].

Interventions to Reduce Outdoor Air Pollution—Environmental and policy actions—such as emission-control measures, urban land use policies, and building codes—are associated with improvements in cardiopulmonary health. Vehicle-related policies, such as school bus electrification or use of cleaner fuel, result in health and environmental benefits as well as cost-savings[187,188]. The institution of low and ultra-low emission zones in London, England, resulted in a reduced incidence of respiratory issues and other long-term health conditions in the region[189]. The alternative transportation program during the 1996 Atlanta Olympics was associated with reductions in particulate matter and O_3, as well as reduced asthma acute care visits and hospitalizations among youth[190]. On a national level, the Clean Air Act has substantially reduced key pollutants despite population and energy growth, contributing to improved child lung function and growth, lower incidence of childhood asthma, and extension of the average life expectancy in the United States[191–196].

Ecological studies following the closure of industrial pollutant sources provide strong evidence for the health benefits of emission reduction. For example, the closure of three coal powered plants in Chicago was associated with a 12% reduction in asthma emergency department visits among children ages 0-4 years in nearby zip codes[197]. Dramatic reductions in adult cardiovascular and respiratory and pediatric asthma emergency department visits (by 20% to >40%) were found in the surrounding area following the closure of a Pittsburgh coal coking operation[198,199], aligned with substantial reductions in SO_2 and coal-related particulate matter. Nationwide, retirement of U.S. coal powered plants has been linked with an average 3.6% reduction in monthly mortality in older adults downwind of the plants compared to control counties (upwind)[200]. Greening initiatives like the Green Heart Project in Louisville suggest likely cardiovascular benefits, including lower levels of systemic inflammatory markers[201,202].

There has been limited direct evaluation of personal interventions to reduce exposure to pollutants while in outdoor spaces, including masking. Modeling studies suggest that the potential health benefits of masking during poor air quality events (e.g., wildfire smoke) depend primarily on the filtration efficiency and fit of the mask or respirator, as well as user compliance[203]. Some observational data suggest modest reductions in lower respiratory symptoms during wildfire smoke events associated with longer use of high-efficiency particulate

air (HEPA) air cleaners and recall of public health advisories, whereas mask use and evacuation were not associated with reduced symptoms[204]. Land use interventions (such as prescribed burns) may reduce the intensity and impact of such events upstream[205,206], but prospective studies are lacking. Overall, these findings demonstrate that environmental policy and planning can function as preventive respiratory and cardiovascular care, though more rigorous prospective and retrospective evaluations are needed to better understand real-world impacts on exposure and cardiopulmonary health.

Interventions to Modify the Biologic Response-- A growing body of research focuses on the relationship between nutrition and lung health. Epidemiological data suggest that higher fruit and vegetable intake is associated with a reduced impact of annual $PM_{2.5}$ exposure on lung function[207], while higher omega-3 intake is associated with a reduced symptomatic response to indoor $PM_{2.5}$ among children with asthma[208] and higher circulating omega-3 is associated with reduced risk of COPD in response to long-term $PM_{2.5}$ and NO_2 exposures[209]. Proposed mechanisms for these effects include: pro-/antioxidant balance; nutrient interaction with the gut microbiome (e.g., fiber), pro-/anti-inflammatory balance (e.g., advanced glycation end-products from meat, omega-3 fatty acids)[210,211], and mechanical or inflammatory effects from obesity, which has also been found to modify respiratory health response to inhaled pollutants[212]. Ongoing work in the United States (Omega-COPD[213]) and Canada (FIRE-Diet[214]) is examining the potential of dietary interventions to counter adverse effects of inhaled exposures.

Further, lung-gut interactions and the possibility of nutritional interventions targeting the microbiome for inhaled exposures—including ones targeted to lung microbiota—are burgeoning scientific topics. The respiratory tract microbiome should be considered as a critical gateway for pollution effects on pulmonary health[215]. Currently, most work on this topic involves pre-clinical models and controlled exposures. However, the beneficial effects of some microbial products (e.g. short chain fatty acids, probiotics) on mucosal barrier function and other immunoregulation[215,216] suggest that microbiome-related intervention strategies may mitigate the consequences of pollutant exposures. However, further mechanistic evidence is needed, with focus on the lung/lower respiratory tract compartments, to inform rational design of therapies for prevention or mitigation.

Conclusion

Workshop participants highlighted the adverse effects of environmental exposures, natural disasters, and extreme weather events on cardiopulmonary health across the lifespan and discussed the state of the science on underlying mechanisms. Emerging tools and refined technologies (e.g., geospatial AI, robotics mimicking human respiration, three-dimensional cell culture and physiologically-relevant models, and lower cost sensors) offer opportunities to advance the field and facilitate the development of mitigation strategies. Although growing knowledge of exposure-related health effects and underlying mechanisms provides a strong foundation for developing interventions, important gaps remain in evaluating the effectiveness, implementation, scalability, and equity of strategies to reduce exposures and improve health. Prior and ongoing studies are examining approaches to cleaner outdoor air (policy, land use management), cleaner indoor air (cleaner fuels in cook stoves, air purifiers, building engineering principles), and mitigation of health effects (diet, microbiome), and additional rigorous intervention and implementation research is needed to identify the most effective and sustainable

solutions. Addressing these critical research gaps will require interdisciplinary teams, attention to multimorbidity, and investment in the data, infrastructure, and translational research bridging discovery to effective interventions to improve health and prevent and treat disease.

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